

The King's Diagonal

Counting Lattice Paths Across the Chessboard

A Pedagogical Introduction to the Delannoy Numbers

Lee McCulloch-James

Department of Natural & Unnatural Sciences

Dwight School London

May 24, 2026

Abstract

We pose a simple chessboard question: in how many ways can a king cross the board diagonally, using only forward moves? From this we develop the full combinatorial machinery of the Delannoy triangle. The exposition is aimed at sixth-form and early undergraduate students. We construct the triangle from a three-parent recurrence, prove a closed binomial formula, verify it against the chessboard count of 48 639, and close with the silver-mean growth rate $3 + 2\sqrt{2}$. No knowledge beyond binomial coefficients is assumed.

1 The Question

Place a king in the corner a1 of an empty chessboard. He is to walk to the diagonally opposite corner h8. To keep things finite and well-defined, we restrict his legal moves to those that make progress toward the goal:

- one square East, one square North, one square North-East (the diagonal).

He may never retreat. *How many distinct walks are available to him?*

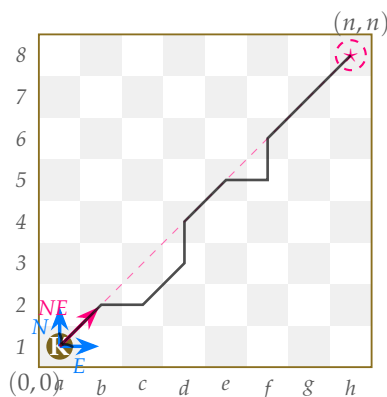


Figure 1: The king walks from a1 to h8 using only East, North, and North-East moves. Three legal first moves are shown in colour; one example completed path is drawn faintly across the board. The dashed line marks the Euclidean geodesic.

The answer though you might suspect small at 48 639, is large enough to surprise but small enough to verify by a hand-built table. Our task is to understand *why* it is that number, and to see the structure that makes the answer compressible into a few lines of arithmetic.

2 The Two-Step Cousin: Pascal's Triangle

Before introducing the diagonal, consider the simpler problem where the king moves only East or North, effectively a rook walking forward.

A rook-walk from $(0,0)$ to (n,k) consists of n North-steps and k East-steps in some order. The number of such walks is the number of ways to choose which k of the $n + k$ moves are East:

$$R(n,k) = \binom{n+k}{k}. \quad (1)$$

For the chessboard corner-to-corner traversal $(7,7)$ this gives $\binom{14}{7} = 3432$.

The values $R(n,k)$ are the entries of Pascal's triangle, traditionally displayed as a centred isocles arrangement so the symmetry is visible:

$$\begin{array}{ccccccc}
 & & & & 1 & & & & \\
 & & & & 1 & & 1 & & \\
 & & & 1 & & 2 & & 1 & \\
 & & 1 & & 3 & & 3 & & 1 \\
 & 1 & & 4 & & 6 & & 4 & & 1 \\
 1 & & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
 & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1
 \end{array}$$

Each entry is the sum of the two immediately above it, encoding the recurrence

$$R(n,k) = R(n-1,k) + R(n,k-1), \quad (2)$$

which says the last step of any walk to (n,k) was either a North-step (arriving from $(n-1,k)$) or an East-step (arriving from $(n,k-1)$). Sum the two predecessor counts; you have the current count.

The central binomial spine

Running down the centre of Pascal's triangle is the sequence

$$\binom{2n}{n} = 1, 2, 6, 20, 70, 252, 924, 3432, 12870, \dots \quad (3)$$

These are the **central binomial coefficients**. Geometrically they count the rook-walks from $(0,0)$ to (n,n) on an $n \times n$ square, because such a walk must interleave exactly n East-steps among n North-steps. The chessboard answer $\binom{14}{7} = 3432$ is the eighth term of this spine, sitting at row $n = 7$. Three properties of this spine matter for what follows.

Closed form. The exact recurrence

$$\binom{2n}{n} = \frac{2(2n-1)}{n} \binom{2(n-1)}{n-1} \quad (4)$$

expresses each entry in terms of its predecessor, which allows the consecutive ratios to be computed directly.

Consecutive ratios. The ratio of one term to the next is

$$\frac{\binom{2n}{n}}{\binom{2(n-1)}{n-1}} = 4 - \frac{2}{n}, \quad (5)$$

which crawls toward 4 from below. Numerically:

n	$\binom{2n}{n}$	ratio	gap to 4	ratio / 4
1	2	2.000	2.000	0.500
2	6	3.000	1.000	0.750
3	20	3.333	0.667	0.833
4	70	3.500	0.500	0.875
5	252	3.600	0.400	0.900
6	924	3.667	0.333	0.917
7	3432	3.714	0.286	0.929

The ratios are settling toward the limit, but the gap to 4 closes only at the rate $2/n$.

Asymptotic. The Stirling approximation gives the leading-order behaviour

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}} \quad (6)$$

which says the spine grows like 4^n divided by a slow $\sqrt{n}$ correction. The 4 is the dominant rate; the $1/\sqrt{n}$ is the reason consecutive ratios approach 4 rather than equal it.

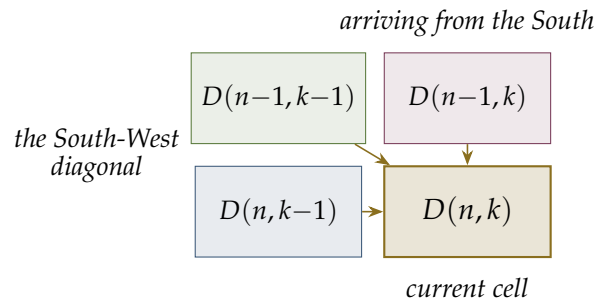
The number 4 deserves a moment of attention. Each rook-step is one of two choices, and reaching (n, n) takes $2n$ steps, so the total number of length- $2n$ walks the rook *could* make (without the constraint of ending at (n, n)) is $2^{2n} = 4^n$. The central binomial coefficient is the fraction of those walks that hit the target. The asymptotic $4^n / \sqrt{\pi n}$ records this exactly: the spine grows like the total walk count, attenuated by the $1/\sqrt{n}$ probability of landing on the diagonal.

So Pascal's spine grows like $2^2 = 4$ because the rook has 2 choices per step and the spine entry sits two indices' worth of growth from the origin.

3 The Three-Parent Rule

Now restore the diagonal step. A walk to (n, k) may terminate in any of *three* final moves:

- East from $(n, k-1)$; North from $(n-1, k)$; North-East from $(n-1, k-1)$.



Each cell is fed by exactly three neighbours and to determine $D(n, k)$ one needs only the three immediately adjacent precomputed cells. The rule is local in that no global information is required, no backtracking. This is the same property that makes Pascal's triangle computable by elementary-school arithmetic, with the addition of one extra parent. Each predecessor contributes its own count of walks to define the **Delannoy numbers** as:

$$D(n, k) = D(n, k-1) + D(n-1, k) + D(n-1, k-1) \quad (7)$$

with the boundary condition $D(0, 0) = 1$ and $D(n, k) = 0$ whenever n or k is negative.

Along the axes, only one walk exists, straight East or straight North, so $D(n, 0) = D(0, k) = 1$. The interior is then determined entirely by (7). Filling cells row by row from the origin gives the following grid:

$n \backslash k$	0	1	2	3	4	5	6	7
0	1	1	1	1	1	1	1	1
1	1	3	5	7	9	11	13	15
2	1	5	13	25	41	61	85	113
3	1	7	25	63	129	231	377	575
4	1	9	41	129	321	681	1289	2241
5	1	11	61	231	681	1683	3653	7183
6	1	13	85	377	1289	3653	8989	19825
7	1	15	113	575	2241	7183	19825	48639

The entry $D(7, 7) = 48\,639$ in the bottom-right corner is our chessboard answer.

We can verify a few cells by hand to confirm the rule.

- $D(1, 1) = D(1, 0) + D(0, 1) + D(0, 0) = 1 + 1 + 1 = 3$. Walks to (1,1): East-then-North, North-then-East, or one diagonal.
- $D(2, 2) = D(2, 1) + D(1, 2) + D(1, 1) = 5 + 5 + 3 = 13$.
- $D(7, 7) = D(7, 6) + D(6, 7) + D(6, 6) = 19825 + 19825 + 8989 = 48\,639$.

4 The Central Delannoy Spine

Following the diagonal $n = k$ down the table yields the **central Delannoy numbers**:

$$D(n, n) = 1, 3, 13, 63, 321, 1683, 8989, 48639, 265729, \dots \quad (8)$$

This is the Delannoy analogue of Pascal's central spine. Each term counts the king-walks across an $n \times n$ board. The chessboard answer $D(7, 7) = 48,639$ sits at row $n = 7$, directly comparable to the rook's $\binom{14}{7} = 3432$.

Noting a few features of the spine will repay our attention.

Divisibility. Every term after the first appears to be divisible by 3:

$$3, 13, 63 = 3 \cdot 21, 321 = 3 \cdot 107, 1683 = 3 \cdot 561, 8989 = ?$$

But at $n = 6$ the divisibility is more subtle: $8989 = 89 \cdot 101$, so the divisibility by 3 already fails. The stronger statement is that $D(n, n) \equiv 1 \pmod{3}$ when n is divisible by 3, and a small modular calculation handles the remaining residues. We mention this only as a flag that the spine carries genuine arithmetic content; we will not pursue the proof here.

Consecutive ratios. As with Pascal's spine, the ratio of consecutive terms is the natural measure of growth. Computing:

n	$D(n, n)$	ratio	gap to $3 + 2\sqrt{2}$	ratio / $(3 + 2\sqrt{2})$
1	3	3.000	2.828	0.515
2	13	4.333	1.495	0.743
3	63	4.846	0.982	0.831
4	321	5.095	0.733	0.874
5	1683	5.243	0.585	0.900
6	8989	5.341	0.487	0.916
7	48639	5.411	0.417	0.928
8	265729	5.463	0.365	0.937

The shape of this convergence is uncannily similar to Pascal's. Both spines approach their asymptotic ratio from below, both close their gap at a rate roughly proportional to $1/n$, and both pass through ratio-over-limit values of 0.93 at $n = 7$. The asymptotic ratio itself differs, but the convergence dynamics do not as both sequences are diagonal extractions of a bivariate generating function, and both have asymptotics of the form

$$a_n \sim \frac{\rho^n}{C\sqrt{n}} \quad (9)$$

where ρ is the dominant growth rate and C is a constant. The $1/\sqrt{n}$ correction is what makes both ratios drag.

Asymptotic. The saddle-point analysis of the bivariate Delannoy generating function (which we treat in the next section) yields

$$D(n, n) \sim \frac{(3 + 2\sqrt{2})^n}{\sqrt{\pi n (3\sqrt{2} - 4)}} \quad (10)$$

with $3 + 2\sqrt{2} = (1 + \sqrt{2})^2$ as the dominant rate. The denominator constant $\sqrt{\pi n (3\sqrt{2} - 4)}$ matches the structural role of the $\sqrt{\pi n}$ in (6).

Side by side. The parallel between the two spines looks thus:

	Spine	Asymptotic growth
Pascal (rook, two choices)	$\binom{2n}{n}$	$4^n / \sqrt{\pi n}$
Delannoy (king, three choices)	$D(n, n)$	$(3 + 2\sqrt{2})^n / \sqrt{\pi n (3\sqrt{2} - 4)}$

The growth rates 4 and $3 + 2\sqrt{2}$ are both *squared expansion factors*: the rook's is $2^2 = 4$, with the 2 coming from its two-choice step, while the king's is $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$, with the $1 + \sqrt{2}$ being the silver mean. The squaring is the same in both cases, because to reach the spine entry at row n one must do n worth of expansion on each of the two axes. What changes is the per-step factor: a rational 2 for the rook, the algebraic silver mean due to the diagonal step for the king. We will see where the silver mean comes from in section 6.

5 A Closed Formula

The recurrence is enough to compute any single value, but it would be more satisfying to have a closed expression. The key observation to achieve this is that any walk to (n, k) can be classified by the number of diagonal steps it contains. Suppose a walk uses exactly j diagonal moves. Each diagonal contributes one unit to both coordinates, so the remaining moves must contribute $k - j$ East-units and $n - j$ North-units. The total number of moves is therefore $j + (n - j) + (k - j) = n + k - j$.

To count walks with exactly j diagonals: choose which j of the $n + k - j$ slots are diagonals $\binom{n+k-j}{j}$, then choose which of the remaining $n + k - 2j$ slots are East-steps $\binom{n+k-2j}{k-j}$. Multiplying and summing over all valid j :

Proposition 1. For $n, k \geq 0$,

$$D(n, k) = \sum_{j=0}^{\min(n, k)} \binom{n+k-j}{j} \binom{n+k-2j}{k-j}. \quad (11)$$

Remark. An equivalent and often more convenient form is

$$D(n, k) = \sum_{j=0}^{\min(n, k)} \binom{n}{j} \binom{k}{j} 2^j,$$

obtained by reorganising the choice as “pick the j diagonals from the n North-units and the k East-units, each diagonal merging one of each.” The factor 2^j tracks the freedom to interleave.

Verification at the chessboard

Apply the formula with $n = k = 7$:

$$D(7, 7) = \sum_{j=0}^7 \binom{7}{j}^2 2^j.$$

The seven terms are tabulated below showing closed formula and recurrence agree.

j	$\binom{7}{j}^2$	2^j	contribution
0	1	1	1
1	49	2	98
2	441	4	1764
3	1225	8	9800
4	1225	16	19600
5	441	32	14112
6	49	64	3136
7	1	128	128
total			48639

6 Where the Silver Mean Comes From

The Pascal spine grows like 4 and the Delannoy spine grows like $3 + 2\sqrt{2}$. Both are squared expansion factors, but the per-step expansion changes from the integer 2 to the silver mean $1 + \sqrt{2}$. We now show why.

The cleanest derivation arrives by *generating functions*. For Pascal, the two-parent recurrence is encoded by

$$\sum_{n,k \geq 0} R(n,k) x^n y^k = \frac{1}{1 - x - y}. \quad (12)$$

The denominator records the two allowed step-monomials, x for North and y for East. Restricting to the diagonal $n = k$ amounts to setting $x = y = z$, which gives $1/(1 - 2z)$, a geometric series with dominant pole at $z = 1/2$. The radius of convergence is therefore $1/2$, and the diagonal coefficients grow like $(1/(1/2))^{2n} = 4^n$. The 4 is the *square* of the inverse radius because each z carries one unit of both indices, so the rate per unit of n is doubled.

For Delannoy, the three-parent recurrence is encoded by

$$\sum_{n,k \geq 0} D(n,k) x^n y^k = \frac{1}{1 - x - y - xy}. \quad (13)$$

The new xy monomial records the diagonal step, which contributes one unit to each index simultaneously. Setting $x = y = z$ gives

$$\frac{1}{1 - 2z - z^2} \quad (14)$$

whose denominator is a *quadratic* in z , not linear as in the Pascal case. The dominant pole lies at the positive root of $z^2 + 2z - 1 = 0$, namely

$$z_* = -1 + \sqrt{2} = \frac{1}{1 + \sqrt{2}}, \quad (15)$$

the reciprocal of the silver mean. The radius of convergence is therefore $\sqrt{2} - 1$, the inverse-radius is $1 + \sqrt{2}$, and squaring gives the spine growth rate

$$(1 + \sqrt{2})^2 = 3 + 2\sqrt{2} \approx 5.828. \quad (16)$$

The silver mean appears precisely because the third (diagonal) step has promoted the characteristic equation from linear to quadratic. Had we added *two* extra steps to Pascal (say NE and NW, or NE and SE), the characteristic equation would have been cubic, and the dominant root would have been a different algebraic number. The silver mean is the fingerprint of the single-diagonal-extension.

A reading of the edge-sum

A natural first guess is that the per-step expansion factor should equal the sum of the available edge lengths. Pascal has two unit edges, so its edge sum is $1 + 1 = 2$, which is indeed the expansion factor. Delannoy has two unit edges and one diagonal of length $\sqrt{2}$, so the natural edge sum is $1 + 1 + \sqrt{2} = 2 + \sqrt{2} \approx 3.414$. But the actual per-step expansion factor is $1 + \sqrt{2} \approx 2.414$, one less than the edge sum.

The discrepancy is the difference between the *trace* and the *dominant eigenvalue* of the step transfer matrix. The transfer matrix is the linear operator that advances the spine by one row. For Pascal we are looking for a number M such that

$$\binom{2n}{n} \sim M \cdot \binom{2(n-1)}{n-1}$$

in the limit. As we computed earlier, the ratio approaches 4. So the transfer matrix for Pascal's spine is the 1×1 scalar

$$T_{\text{Pascal}} = [4].$$

A scalar has just one eigenvalue, equal to itself, and a trace equal to itself. So the dominant eigenvalue and the trace are both 4. They coincide trivially.

For Delannoy the situation is different because the spine recurrence involves *two* previous terms, not one. The diagonal step links cells whose indices differ by $(1, 1)$, which on the spine means the diagonal step connects $D(n, n)$ back to $D(n-1, n-1)$, the immediate predecessor. But the orthogonal steps each link $D(n, n)$ back to cells *off* the spine, which we have to express in terms of spine entries two steps back. The result is that the spine sequence approximately satisfies the second-order recurrence

$$D(n, n) \approx 6D(n-1, n-1) - D(n-2, n-2) \quad (17)$$

to leading order. (We met this recurrence above in the form $a_n = 6a_{n-1} - a_{n-2}$. It is exact for the related Pell numbers and the balancing numbers; for $D(n, n)$ itself it captures the dominant growth but not the full structure.) A second-order recurrence is run by a 2×2 transfer matrix.

Write the state vector as the pair $\begin{pmatrix} D(n, n) \\ D(n-1, n-1) \end{pmatrix}$. Then (17) becomes

$$\begin{pmatrix} D(n+1, n+1) \\ D(n, n) \end{pmatrix} = T_{\text{Delannoy}} \begin{pmatrix} D(n, n) \\ D(n-1, n-1) \end{pmatrix}, \quad T_{\text{Delannoy}} = \begin{pmatrix} 6 & -1 \\ 1 & 0 \end{pmatrix}.$$

The top row encodes the recurrence; the bottom row simply shifts the old top entry down. This 2×2 matrix has trace

$$\text{tr } T_{\text{Delannoy}} = 6 + 0 = 6$$

and determinant

$$\det T_{\text{Delannoy}} = 6 \cdot 0 - (-1) \cdot 1 = 1.$$

Its characteristic polynomial is therefore $\lambda^2 - 6\lambda + 1 = 0$, the same polynomial we encountered before, with roots

$$\lambda_{\pm} = 3 \pm 2\sqrt{2} = (1 \pm \sqrt{2})^2.$$

The dominant eigenvalue is $\lambda_+ = 3 + 2\sqrt{2} \approx 5.828$, which is the spine growth rate. The trace is 6, which is *not* the spine growth rate.

For a 2×2 matrix the trace equals the *sum* of the two eigenvalues, $\lambda_+ + \lambda_- = (3 + 2\sqrt{2}) + (3 - 2\sqrt{2}) = 6$. The dominant eigenvalue is only one of these two contributions. For Pascal the matrix was 1×1 so there was only one eigenvalue to be confused with the trace; for Delannoy the matrix is 2×2 and the trace exceeds the spine growth rate by exactly the sub-dominant eigenvalue $\lambda_- = 3 - 2\sqrt{2} \approx 0.172$. This sub-dominant eigenvalue is small but non-zero. It represents a *decaying mode*: a component of the spine sequence that shrinks as n grows. The actual spine ratio approaches the dominant eigenvalue from below precisely because the sub-dominant mode is contributing a small negative correction that vanishes in the limit. We saw this already in the table of consecutive ratios.

On a per-lattice-step basis (taking square roots, since the spine grows n at a time but each spine step represents two lattice steps), the two modes are

$$\sqrt{\lambda_+} = 1 + \sqrt{2} \approx 2.414, \quad \sqrt{\lambda_-} = \sqrt{2} - 1 \approx 0.414.$$

Their sum is $2\sqrt{2} \approx 2.828$, and neither equals the geometric edge sum $2 + \sqrt{2} \approx 3.414$. The edge sum exceeds the dominant mode by exactly 1, and this gap has a clean interpretation: it is the difference between counting the diagonal step by its *Euclidean length* $\sqrt{2}$ and counting it by its *coordinate advance* of 1 per axis. The lattice's spine growth listens to the algebra of the characteristic equation, not to either counting convention directly, and the dominant eigenvalue $1 + \sqrt{2}$ is what the algebra delivers.

There is a further coincidence worth flagging. The $\sqrt{2}$ appearing in the expansion factor is numerically the same as the $\sqrt{2}$ of the diagonal's Euclidean length, but the two arrive by different routes. The lattice $\sqrt{2}$ comes from the discriminant of the characteristic quadratic $z^2 + 2z - 1 = 0$: solving by the quadratic formula gives $z = -1 \pm \sqrt{(2/2)^2 + 1} = -1 \pm \sqrt{2}$, and the $\sqrt{2}$ is the square root of the sum $1 + 1$ of squared coefficients in the characteristic equation. The geometric $\sqrt{2}$ comes from Pythagoras applied to a unit square: the diagonal of a unit square has length $\sqrt{1^2 + 1^2} = \sqrt{2}$.

That these two square-roots-of-two coincide is a consequence of the *symmetry* of the Delannoy diagonal step, which advances $(1, 1)$, one unit on each axis. If we modified the lattice so that the diagonal step advanced $(1, 2)$ instead (a knight-like jump), the characteristic equation would become cubic and its dominant root would be an algebraic number unrelated to the Euclidean length $\sqrt{1^2 + 2^2} = \sqrt{5}$ of that step. The edge-sum reading would predict expansion factor $1 + 1 + \sqrt{5} \approx 4.236$, whereas the actual growth rate would be $1/z_*$ where z_* is the smallest positive root of $z^3 + 2z - 1 = 0$, giving approximately 2.206. The two quantities would differ by a factor of nearly two and would not even be elements of the same number field.

So the lattice's silver mean is an algebraic object that the geometric edge-sum picture only partly anticipates. The Delannoy case is the clean one because its diagonal advances symmetrically; in any asymmetric variant, the geometric and algebraic readings would part company entirely. The square-symmetry of the chessboard is doing quiet work in the background, ensuring that Pythagoras and the quadratic formula deliver the same number.

The Pell-recurrence skeleton

The silver mean satisfies $\alpha^2 = 2\alpha + 1$, the defining equation of the silver-mean family. This is the algebraic shadow of the recurrence

$$a_n = 6a_{n-1} - a_{n-2}, \tag{18}$$

whose characteristic polynomial $\lambda^2 - 6\lambda + 1$ has roots $3 \pm 2\sqrt{2} = (1 \pm \sqrt{2})^2$. The central Delannoy numbers do not quite satisfy this recurrence exactly (the true Delannoy recurrence has more structure), but they share its dominant eigenvalue. The same recurrence governs the Pell numbers $1, 2, 5, 12, 29, 70, \dots$, the silver-mean continued-fraction convergents, and the balancing numbers of Behera and Panda. The Delannoy spine sits inside this *Pell family* the way the central binomial coefficients sit inside the Pascal family. The diagonal step is what promotes the structure from Pascal to Pell, from gold to silver, from a rational growth rate to an algebraic one.

7 Why So Much Larger Than Pascal?

The rook-only walk count is $\binom{14}{7} = 3432$. The king's count is 48 639. The ratio is

$$\frac{48\,639}{3432} \approx 14.2.$$

The diagonal step compounds as every interior cell has three predecessors instead of two, and the counts feed forward multiplicatively. By row $n = 20$ the king's count exceeds the rook's by a factor of order 10^7 . By $n = 50$ the gap is astronomical.

This compounding is exactly the difference between the growth rates 4 and 5.83 raised to large powers. For large n ,

$$\frac{D(n,n)}{\binom{2n}{n}} \sim \left(\frac{3 + 2\sqrt{2}}{4} \right)^n \cdot C \approx 1.457^n \cdot C \quad (19)$$

where C is a constant of order one absorbing the $\sqrt{n}$ corrections. Each step compounds the king's lead over the rook by about 46%.

8 Closing Thought

The chessboard problem has the cleanest possible answer: *three parents, one rule, build the table*. Every cell is just the local sum of its three predecessors, and the global count emerges by recursion. The same logic gives $D(2,2) = 13$ by hand and $D(7,7) = 48\,639$ by table, with no algebraic trickery.

What makes the Delannoy numbers worth attention beyond this exercise is the arithmetic that grows on top of the geometry: the central values 1, 3, 13, 63, 321, 1683, ... also count king-walks across a chessboard, appear as evaluations of Legendre polynomials at the value 3, satisfy non-trivial congruences modulo primes, and share their generating function with the Schröder numbers from one-dimensional walk theory.

But all of that begins with the same observation we made in section 7: three parents, one rule. The rest is just following the consequences.

Exercise. A bishop on an empty board moves only diagonally. If we allow the bishop to step North-East, South-East, or stay diagonal in some sense, find a sensible variant of the question. What triangle does the bishop's walk generate? *Hint: think about parity colouring.*

Exercise. Compute $D(10,10)$ in two ways: by extending the table, and by Proposition 1. Confirm both give 8 097 453.

Exercise. Show directly from the closed formula that $D(n,k) = D(k,n)$. What property of king-walks does this symmetry reflect?

Exercise. Verify by hand that the consecutive-ratio formula $\binom{2n}{n} / \binom{2(n-1)}{n-1} = 4 - 2/n$ holds at $n = 5$. Then compute the corresponding ratio $D(5,5)/D(4,4)$ and observe that it does not simplify to a clean rational of the form $a - b/n$, despite approaching the same kind of limit. What does this tell you about the difference between the two spines?